



BITS Pilani

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CE F231 Fluid Mechanics

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Lecture 29: Notches and Weirs

Learning Outcome

- What are notches and weirs (and their classifications)?
- Estimation of flow over different types of notches and weirs
- Error in estimating discharge over notches or weirs due to error in measuring head
- Time required to empty tank with rectangular/triangular notch
- Estimation of discharge accounting for velocity of approach
- Estimation of discharge using empirical equations
- Discharge over broad-crested weir and drowned weir

Notches

- Opening provided in the side of a tank such that the liquid surface is below the top edge of opening
- Notches made of metallic plates are provided in narrow channels to measure liquid flow rate



Notches

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- Notches made of metallic plates are provided in narrow channels to measure liquid flow rate



https://www.encardio.com/wp-content/uploads/2020/07/ESM-10_V-Notch-for-seepage.pdf

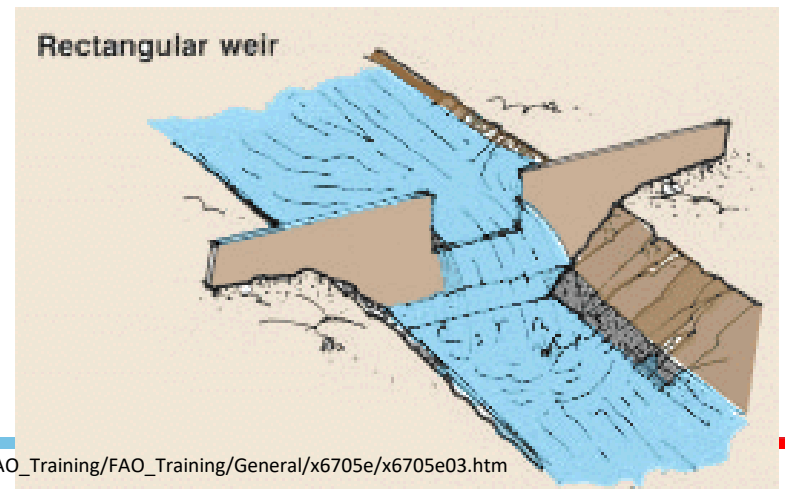
Weirs



- Concrete or masonry structures built across streams or rivers
- Raise water level on upstream side and allow excess water to flow over the weir
- Similar to small dam
- Measure rate of flow in rivers or streams
- Weirs are larger than notches

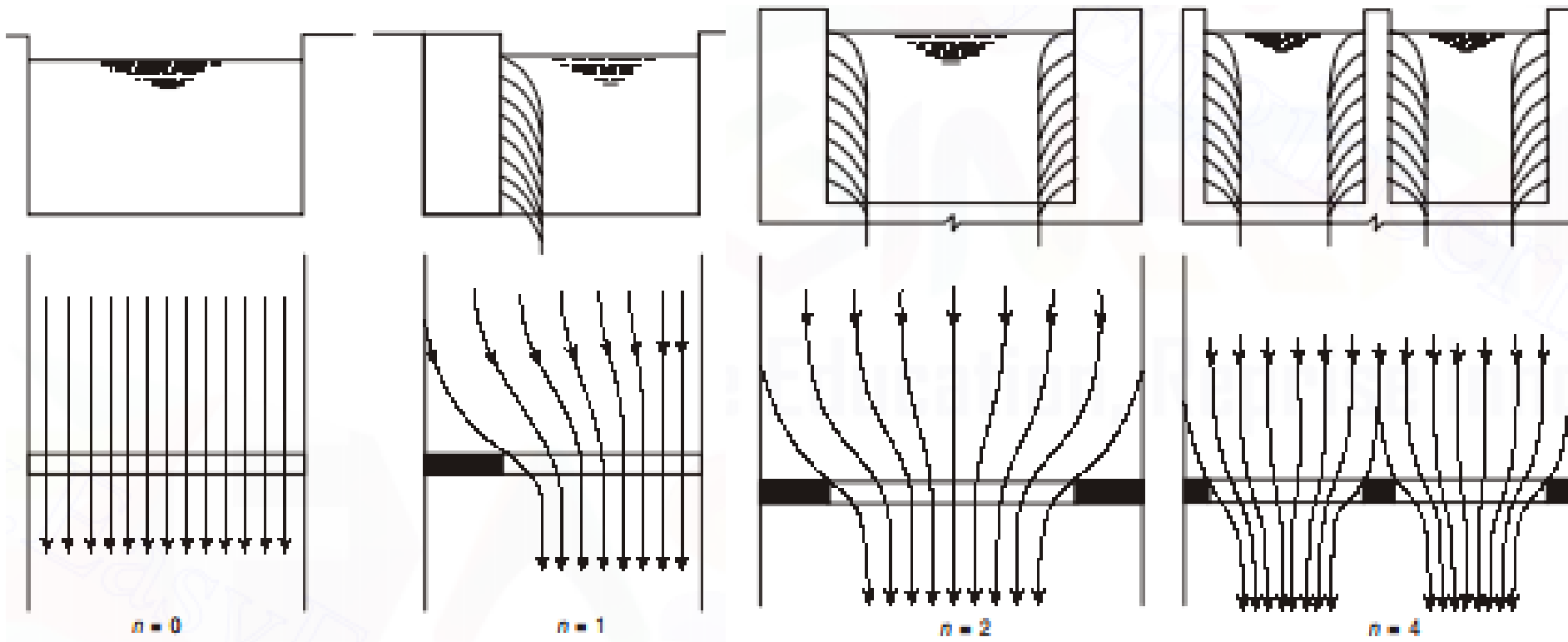


<https://studentlesson.com/notch-and-weir-definition-applications-function-components-diagram-classifications-types-working-principles/>



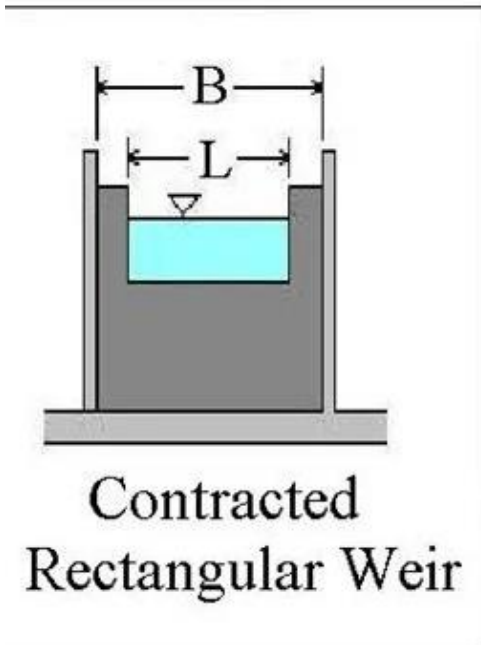
https://www.fao.org/fishery/docs/CDrom/FAO_Training/FAO_Training/General/x6705e/x6705e03.htm

Rectangular weir (suppressed vs contracted)



Rectangular weir with end contraction

- Length of weir $<$ width of channel $\rightarrow$ nappe (sheet of water) contracts
- This type of weir is a contracted weir

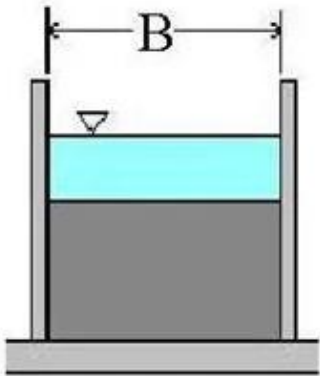


Rectangular weir with end contraction



Suppressed rectangular weir

- Sides of the flow channel act as the ends of a rectangular weir, no side contraction exists, and the nappe does not contract from the width of the channel.
- This type of weir is a suppressed weir



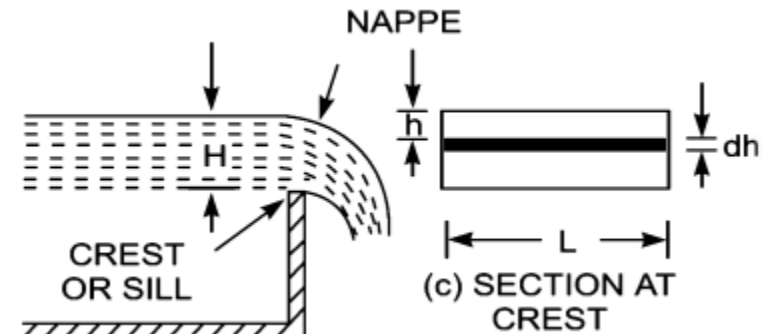
Suppressed
Rectangular Weir



Discharge over suppressed rectangular notch or weir

H = Head of water over the crest

L = Length of the notch or weir



$$dQ = C_d \times \text{Area of strip} \times \text{Theoretical velocity}$$

$$= C_d \times L \times dh \times \sqrt{2gh} \quad \text{(Velocity of approach = 0)}$$

$$Q = \int_0^H C_d \cdot L \cdot \sqrt{2gh} \cdot dh = C_d \times L \times \sqrt{2g} \int_0^H h^{1/2} dh$$

$$= C_d \times L \times \sqrt{2g} \left[\frac{h^{1/2+1}}{\frac{1}{2}+1} \right]_0^H = C_d \times L \times \sqrt{2g} \left[\frac{h^{3/2}}{3/2} \right]_0^H$$

$$= \frac{2}{3} C_d \times L \times \sqrt{2g} [H]^{3/2}.$$

Discharge over triangular notch or weir



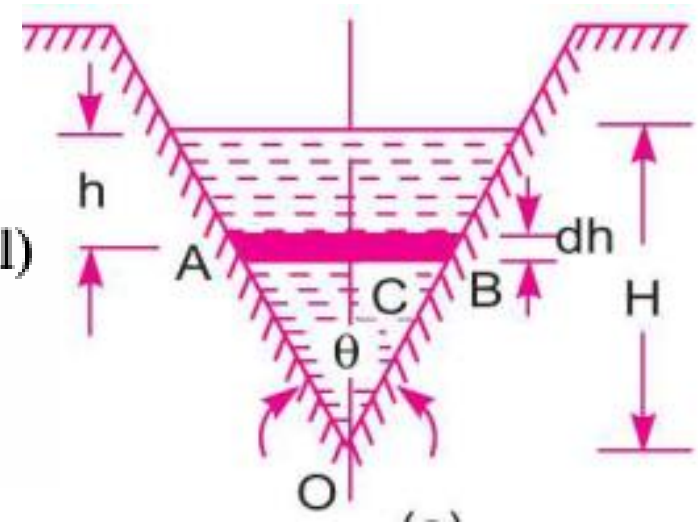
H = head of water above the V- notch

θ = angle of notch

$$dQ = C_d \times \text{Area of strip} \times \text{Velocity (theoretical)}$$

$$= C_d \times 2 (H - h) \tan \frac{\theta}{2} \times dh \times \sqrt{2gh}$$

$$Q = \frac{8}{15} C_d \times \tan \frac{\theta}{2} \times \sqrt{2g} \times H^{5/2}$$



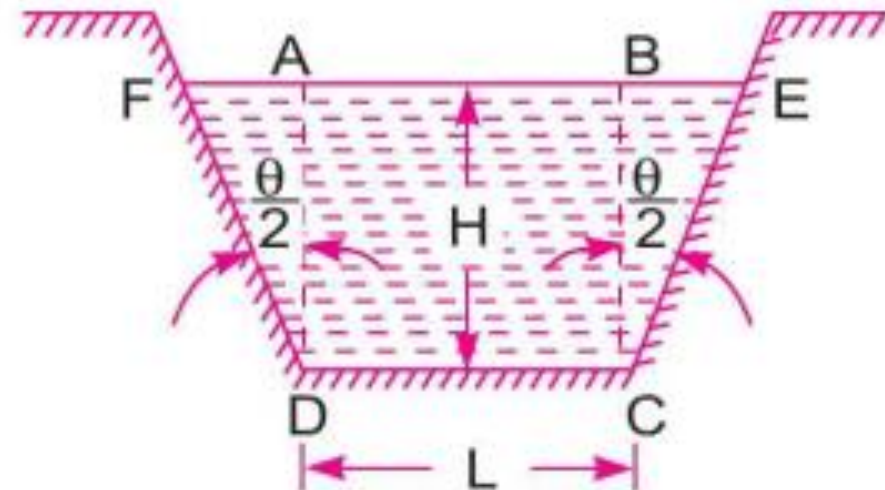
Discharge over trapezoidal notch or weir

innovate

achieve

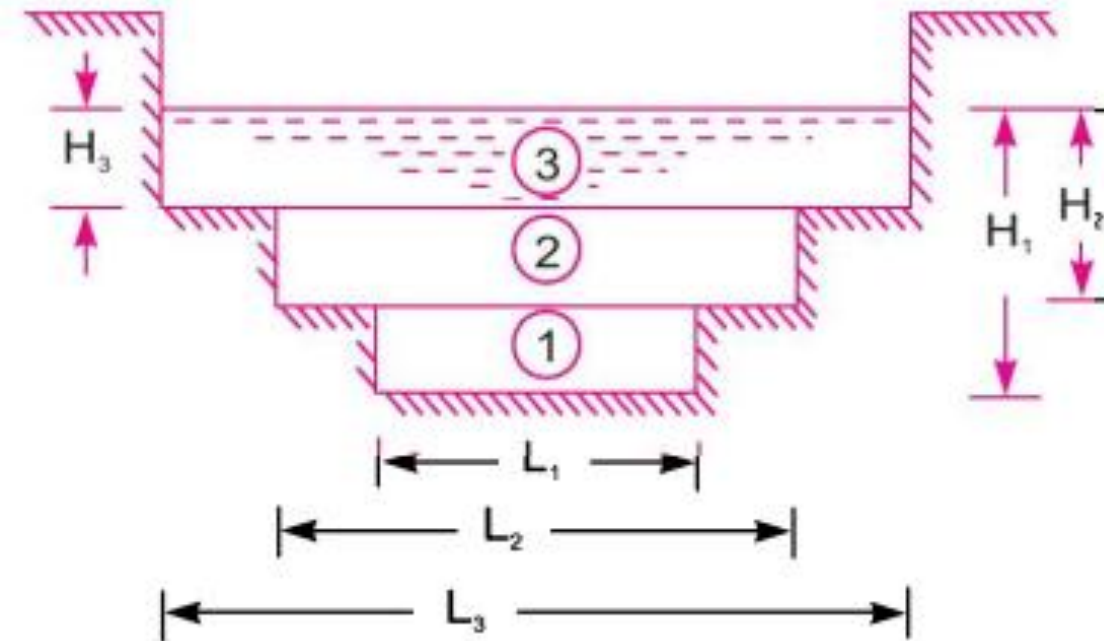
lead

H = Height of water over the notch
 L = Length of the crest of the notch



$$Q = \frac{2}{3} C_{d_1} L \sqrt{2g} \times H^{3/2} + \frac{8}{15} C_{d_2} \times \tan \theta/2 \times \sqrt{2g} \times H^{5/2}$$

Discharge over stepped notch or weir



Q for rectangular weir

$$Q = \int_0^H C_d \cdot L \cdot \sqrt{2gh} \cdot dh$$

$$Q = \frac{2}{3} \times C_d \times L_1 \times \sqrt{2g} [H_1^{3/2} - H_2^{3/2}] \\ + \frac{2}{3} C_d \times L_2 \times \sqrt{2g} [H_2^{3/2} - H_3^{3/2}] + \frac{2}{3} C_d \times L_3 \times \sqrt{2g} \times H_3^{3/2}.$$

Effect of error in measuring head on discharge



For rectangular notch or weir

$$Q = \frac{2}{3} \times C_d \times L \times \sqrt{2g} \times H^{3/2}$$
$$= KH^{3/2} *$$

$$dQ = K \times \frac{3}{2} H^{1/2} dH$$

$$\frac{dQ}{Q} = \frac{K \times \frac{3}{2} \times H^{1/2} dH}{KH^{3/2}} = \frac{3}{2} \frac{dH}{H}$$

1% error in measuring H $\rightarrow$ 1.5% error in measuring Q

Effect of error in measuring head on discharge



For triangular notch or weir

$$Q = \frac{8}{15} C_d \tan \frac{\theta}{2} \sqrt{2g} \times H^{5/2}$$
$$= KH^{5/2}$$

$$dQ = K \frac{5}{2} H^{3/2} \times dH$$

$$\frac{dQ}{Q} = \frac{K \frac{5}{2} H^{3/2} dH}{KH^{5/2}} = \frac{5}{2} \frac{dH}{H}$$

1% error in measuring H $\rightarrow$ 2.5% error in measuring Q

Time required to empty a reservoir with a rectangular notch



$$-Adh = Q \times dT$$

$$Q = \frac{2}{3} C_d \times L \times \sqrt{2g} \times h^{3/2}$$

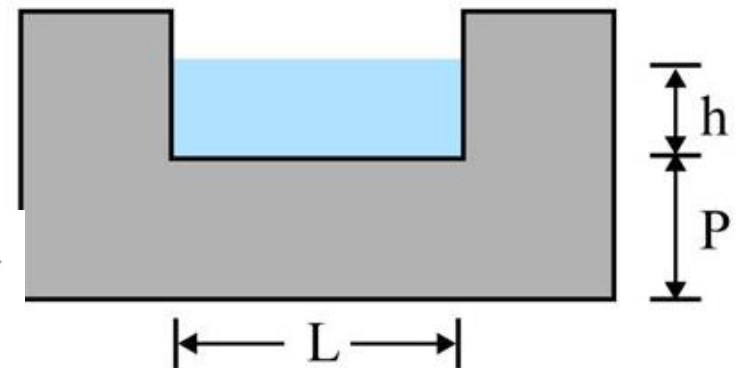
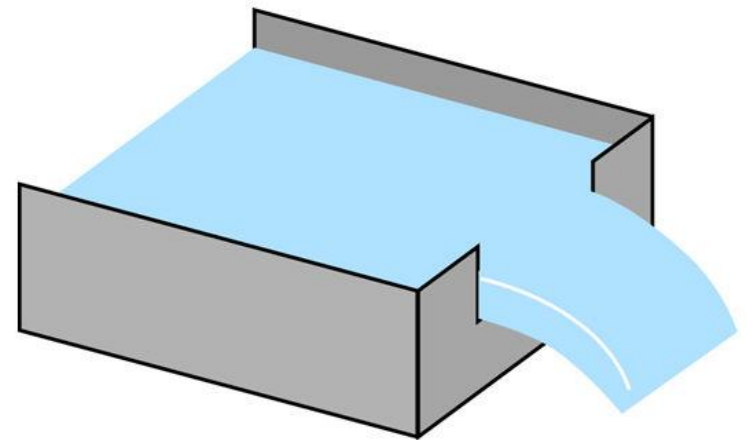
$$dT = \frac{-Adh}{\frac{2}{3} C_d \times L \times \sqrt{2g} \times h^{3/2}}$$

$$T = \frac{3A}{C_d \times L \times \sqrt{2g}} \left[\frac{1}{\sqrt{H_2}} - \frac{1}{\sqrt{H_1}} \right]$$

A = Area of tank

H_1 = Initial height of liquid above the crest of notch

H_2 = Final height of liquid above the crest of notch



Time required to empty a reservoir with a triangular notch



$$-Adh = \dot{Q} \times dT \quad Q = \frac{8}{15} \times C_d \times \tan \frac{\theta}{2} \sqrt{2g} \times h^{5/2}$$

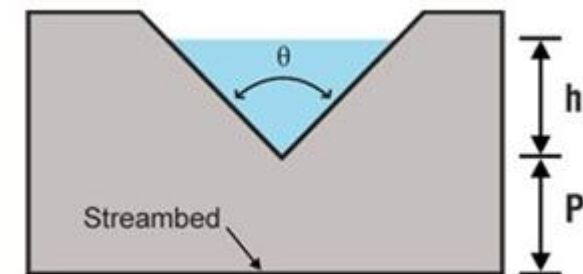
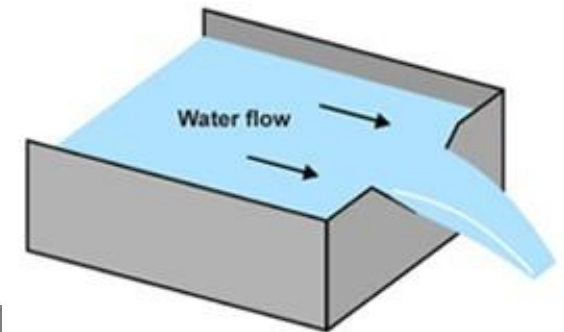
$$dT = \frac{Adh}{\frac{8}{15} \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g} \times h^{5/2}}$$

$$T = \frac{5A}{4 \times C_d \times \tan \frac{\theta}{2} \times \sqrt{2g}} \left[\frac{1}{H_2^{3/2}} - \frac{1}{H_1^{3/2}} \right]$$

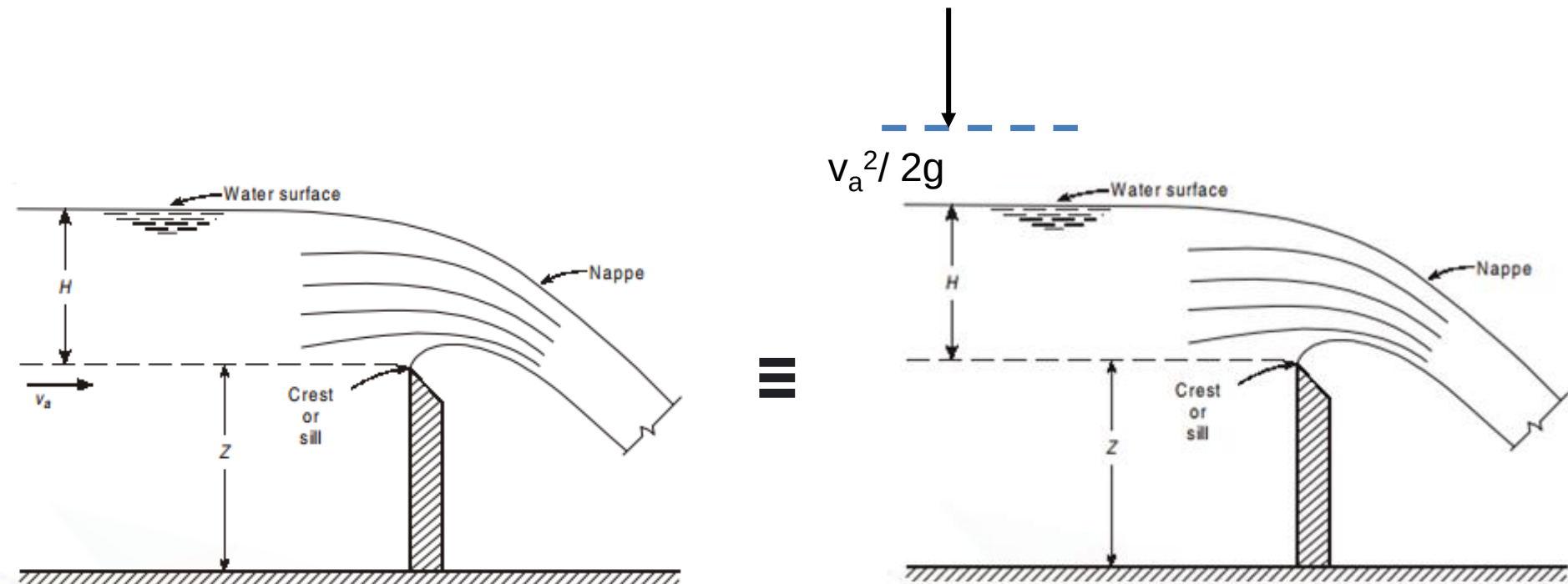
A = Area of tank

H_1 = Initial height of liquid above the crest of notch

H_2 = Final height of liquid above the crest of notch



Discharge accounting for velocity of approach



$$H + \frac{v_a^2}{2g} = \text{Still water head}$$



THANK YOU